

2-Month Generative AI Course Plan (Beginner → Advanced)

Month 1 — Foundations + Deep Learning + NLP + Transformers

WEEK 1 — Python + ML + Math (Fast Revision)

Topics:

- Python basics
- NumPy, Pandas
- Matplotlib
- ML basics
- Loss functions
- Gradient Descent
- Train-test split

Practical:

- Linear regression implementation
- Simple classifier

WEEK 2 — Deep Learning Basics (PyTorch)

Topics:

- Neural networks
- Activation functions
- Optimizers
- Overfitting & regularization
- PyTorch model building

Practical:

- Build NN in PyTorch
- Train on MNIST

WEEK 3 — NLP Fundamentals

Topics:

- Tokenization
- Stemming, Lemmatization
- TF-IDF
- Word embeddings

Practical:

- Sentiment analysis
- Spam detection

WEEK 4 — Attention + Transformers

Topics:

- RNN issues
- Attention mechanism
- Self-attention
- Multi-head attention
- Positional encoding
- Transformer architecture

Practical:

- Build self-attention layer
- Mini transformer

Month 2 — Generative Models, LLMs, RAG, Fine-tuning, Deployment

WEEK 5 — Large Language Models (LLMs)

Topics:

- GPT architecture
- BERT vs GPT
- Pretraining vs Fine-tuning
- LoRA, QLoRA

Practical:

- Load Llama/Mistral
- Generate text

WEEK 6 — Generative Models

Topics:

- Autoencoders
- Variational Autoencoders (VAE)
- GANs
- Diffusion models

Practical:

- Train VAE
- Train small GAN
- Use Stable Diffusion

WEEK 7 — Fine-Tuning + RAG

Topics:

- LoRA fine-tuning
- Vector databases
- Embeddings
- RAG architecture

Practical:

- Fine-tune Llama
- Build RAG chatbot

WEEK 8 — Deployment + Final Projects

Topics:

- FastAPI / Gradio
- GPU usage
- Model quantization
- HuggingFace Spaces

Final Projects:

- AI chatbot
- RAG Q&A; system
- Image generator
- Voice cloning
- AI resume analyzer